Chapter 1 Classical Algebra

Exercise 5

Let $S = \{p + q\alpha + r\alpha^2 \mid p, q, r \in \mathbb{Q}\}$. We first prove that S is a subring of $\mathbb{C}$.

Note that $1 = 1+0\alpha+0\alpha^2 \in S$. Let $p+q\alpha+r\alpha^2, p'+q'\alpha+r'\alpha^2 \in S$, where $p, q, r, p', q', r' \in \mathbb{Q}$. Then we have $(p+q\alpha+r\alpha^2)(p'+q'\alpha+r'\alpha^2) = (p+p')+(q+q')\alpha+(r+r')\alpha^2 \in S$, $-(p+q\alpha+r\alpha^2) = (-p)+(-q)\alpha+(-r)\alpha^2 \in S$ and $(p+q\alpha+r\alpha^2)(p'+q'\alpha+r'\alpha^2) = (pp'+2rq')+(pq'+p'q+2rr')\alpha+(pr'+qq'+qr'+rp')\alpha^2 \in S$. By the subring test, S is a ring.

Next, we construct an inverse explicitly to show that S is a subfield. Let $x = p+q\alpha+r\alpha^2 \in S\setminus\{0\}$, where $p, q, r \in \mathbb{Q}$. Then we have $x^2 = (p^2+4qr)+(2pq+2r^2)\alpha+(2pr+q^2)\alpha^2$ and $x^3 = (p^3+12pqr+2q^3+4r^3)+(3p^2q+6pr^2+6q^2r)\alpha+(3p^2r+3pq^2+6qr^2)\alpha^2$. Let $x^3+k_2x^2+k_1x+k_0 = 0$. Denote the coefficient of α^j in x^i by x_{ij}. Then we have $(x_{30}+x_{31}\alpha+x_{32}\alpha^2)+k_2(x_{20}+x_{21}\alpha+x_{22}\alpha^2)+k_1(p+q\alpha+r\alpha^2)+k_0 = 0$, that is, $(x_{30} + x_{20}k_2 + pk_1 + k_0) + (x_{31} + x_{21}k_2 + qk_1)\alpha + (x_{32} + x_{22}k_2 + rk_1)\alpha^2 = 0$. Since $1, \alpha, \alpha^2$ are linearly independent, comparing coefficients gives us the linear system

$$\begin{cases} k_0 + pk_1 + x_{20}k_2 = -x_{30} \\ qk_1 + x_{21}k_2 = -x_{31} \\ rk_1 + x_{22}k_2 = -x_{32} \end{cases}.$$

We can use the following python code to solve this system.

```python
import sympy as sp

k0, k1, k2 = sp.symbols('k0 k1 k2')
p, q, r = sp.symbols('p q r')

x30 = p**3 + 12 * p * q * r + 2 * (q**3) + 4 * (r**3)
x31 = 3 * (p**2) * q + 6 * p * (r**2) + 6 * (q**2) * r
x32 = 3 * (p**2) * r + 3 * p * (q**2) + 6 * q * (r**2)
x20 = p**2 + 4 * q * r
x21 = 2 * p * q + 2 * (r**2)
x22 = 2 * p * r + q**2

M = sp.Matrix([
    [1, p, x20, - x30],
    [0, q, x21, - x31],
    [0, r, x22, - x32]
])

solution = sp.solve_linear_system(M, k0, k1, k2)

for var in [k0, k1, k2]:
    print(f"{var} =")
    sp.pprint(sp.simplify(solution[var]))
```

The solution is $k_0 = -p^3+6pqr-2q^3-4r^3$, $k_1 = 3p^2-6qr$ and $k_2 = -3p$. Since $x^3+k_2x^2+k_1x+k_0 = 0$, we have $x(x^2+k_2x+k_1+k_0x^{-1}) = 0$. Recall that $\mathbb{C}$ is a field and hence a domain, and $x \neq 0$. Therefore, $x^2 + k_2x + k_1 + k_0x^{-1} = 0$, that is,

$$x^{-1} = -\frac{1}{k_0}x^2 - \frac{k_2}{k_0}x - \frac{k_1}{k_0}.$$

Since x and x^2 are linear combinations of $1, \alpha, \alpha^2$, x^{-1} is also a linear combination of $1, \alpha, \alpha^2$. $\square$

Exercise 2

Assume for contradiction that there exists $a, b \in \mathbb{Z}$ with $b \neq 0$ such that $\left(\frac{a}{b}\right)^2 = 2$. If a is negative and b is positive, then replacing a with $-a$ gives us $\left(\frac{-a}{b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $-a, b$ is also a valid pair of integers satisfying the conditions. If b is negative and a is positive, then replacing b with $-b$ gives us $\left(\frac{a}{-b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $a, -b$ is also a valid pair of integers satisfying the conditions. If both a and b are negative, then replacing a with $-a$ and b with $-b$ gives us $\left(\frac{-a}{-b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $-a, -b$ is also a valid pair of integers satisfying the conditions. Hence, we can assume without loss of generality that $a, b > 0$.

Since we assume that b only takes values in $\mathbb{N}$, by the well-ordering principle, there exists a smallest b satisfying the conditions. Note that

$$\begin{aligned}
\left(\frac{2b-a}{a-b}\right)^2 &= \frac{4b^2-4ab+a^2}{a^2-2ab+b^2} \\
&= \frac{4-4\left(\frac{a}{b}\right)+\left(\frac{a}{b}\right)^2}{\left(\frac{a}{b}\right)^2-2\left(\frac{a}{b}\right)+1} \\
&= \frac{4-4\left(\frac{a}{b}\right)+2}{2-2\left(\frac{a}{b}\right)+1} \\
&= \frac{6-4\left(\frac{a}{b}\right)}{3-2\left(\frac{a}{b}\right)} \\
&= \frac{6-4\left(\frac{a}{b}\right)}{3-2\left(\frac{a}{b}\right)} \times \frac{3+2\left(\frac{a}{b}\right)}{3+2\left(\frac{a}{b}\right)} \\
&= \frac{18+12\left(\frac{a}{b}\right)-12\left(\frac{a}{b}\right)-8\left(\frac{a}{b}\right)^2}{9-4\left(\frac{a}{b}\right)^2} \\
&= \frac{18-8\times 2}{9-4\times 2} \\
&= 2.
\end{aligned}$$

Recall that $\left(\frac{a}{b}\right)^2 = 2$, so we have $a^2 = 2b^2$ and thus $2b^2 - a^2 = 0$. Note that

$$\begin{aligned}
(2b-a)(2b+a) &= 4b^2 - a^2 \\
&= 2b^2 + (2b^2 - a^2) \\
&= 2b^2 \\
&> 0
\end{aligned}$$

since $b > 0$ by assumption. Since $a, b > 0$, we have $2b + a > 0$. Hence, $2b - a > 0$. Also, from $a^2 = 2b^2$ we have $a^2 - b^2 = b^2$. Considering $b > 0$, we have

$$\begin{aligned}
b^2 &> 0 \\
a^2 - b^2 &> 0 \\
(a-b)(a+b) &> 0 \\
a - b &> 0
\end{aligned}$$

since $a+b > 0$. From $2b-a > 0$ and $a-b > 0$ we know that $2b-a, a-b$ is also a pair of positive integers satisfying the conditions. However, from $2b - a > 0$ we have $a < 2b$, and hence $a - b < 2b - b = b$, contradicting the minimality of b. $\square$